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21 June 1983

SUPERSEDING
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22 August 1975

MILITARY SPECIFICATION

PUMP, CENTRIFUGAL: PETROLEUM-PIPELINE, DIESEL-ENGINE-DRIVEN, SKID-MOUNTED; 2-STAGE, 6-INCH, 500-GPM AT 555-FOOT HEAD TO 1,400-GPM AT 275-FOOT HEAD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a skid-mounted, direct-connected, diesel-engine-driven, horizontally split case, centrifugal pump (hereinafter referred to as "pump unit") complete with two 6-inch, 250-pound, flanged-to-grooved-type adapters; an external crossover assembly for series operation; two by 6-inch Y-connections for parallel operation; and all necessary accessories for use in petroleum pipelines.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4320

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-35	- Passivation Treatments for Corrosion Resisting Steel.
UU-T-81	- Tags, Shipping and Stock.
VV-F-800	- Fuel Oil, Diesel.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-P-40	- Packaging and Packing of Hand Tools.

MILITARY

MIL-C-104	- Crate, Wood, Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173	- Varnish, Moisture and Fungus Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-R-196	- Repair Parts, Accessories, and Kits, Mechanical Packaging of.
MIL-P-514	- Plate, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-C-3774	- Crate, Wood, Open 12,000 and 16,000 Pound Capacity.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-M-17194	- Metal, Expanded, Steel.
MIL-T-22085	- Tapes, Adhesive, Preservation and Sealing.

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MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty Single Package.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements For The Control of Electromagnetic Interference.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-1400	- Engine, Gasoline or Diesel, Methods of Test.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Material.
MS 35829	- Hinge, Butt, Continuous (Piano), .075 Thickness, .1878, .250 Pin Dia.
MS 51389	- Raincap, for Air Cleaner Intake, for Industrial Engines.

2.1.2 Government documents and drawings. The following Government document and drawings form a part of this specification to the extent specified herein.

U.S. DEPARTMENT OF AGRICULTURE

Forest Service Standard 5100-1A

DRAWINGS

ME

TA13207E9323	- Pump, Centrifugal, Petroleum Pipeline, DED, Skid-Mounted, 2-Stage, 6-Inch, Assembly.

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(Copies of specifications, standards, document, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Steel Products Manual.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York, NY 10017.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI/ASME - Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

ANSI/AWS - D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A276 - Stainless and Heat Resisting Steel Bars and Shapes.

ASTM A582 - Free Machining Stainless and Heat Resisting Steel Bars, Hot-Rolled or Cold-Finished.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 102 - Photochemically Reactive Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 East Flair Drive, El Monte, CA 91731.)

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HYDRAULIC INSTITUTE (HI)

Rating Standards and Test Code Standards.

(Application for copies should be addressed to the Hydraulic Institute, 712 Lakewood Center North, Cleveland, OH 44107.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15086.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The pump unit shall be in accordance with Top Assembly TA13207E9323 and as specified herein, except that the wiring diagram, control panel assembly, and engine drawings shall not be applicable.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the pump unit. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. When specified, (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein and as shown on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.11).

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3.3.1 Material deterioration and control. The pump unit shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar Metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.4 Environmental.

3.4.1 Operating temperature. The pump unit shall perform as specified herein in any ambient temperature from plus 110° F to minus 25° F.

3.4.2 Storage temperature. The pump unit shall not be damaged by storage in any ambient temperature from plus 145° F to minus 28° F.

3.5 Safety. All shafts or rotating parts of the pumping unit shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces of the equipment, including exhaust pipes exposed to contact by personnel or which create a fire hazard shall be fully guarded or insulated.

3.6 Human factors. The pump unit shall conform to the human factors engineering criteria as specified in MIL-STD-1472.

3.6.1 Noise limits. The noise of the pump unit shall not exceed the limits specified in MIL-STD-1474, category D for steady-state noise when tested in accordance with 4.5.2.5.1.1. The sound level shall comply with the exterior noise requirements of MIL-STD-1474 when tested in accordance with 4.5.2.5.1.2 (aural nondectectability and impulse noise do not apply). The precautions of MIL-STD-1474 (see 6.10) shall be provided if the sound level exceeds 85 dB(A) when tested in accordance with 4.5.2.5.1.3. In addition the distance from the pump unit at which a limit of 85 dBA is met shall be shown on hazard signs mounted on all four sides of each production unit stating: "CAUTION, HEARING PROTECTION REQUIRED WITHIN _____ FEET WHEN EQUIPMENT IS OPERATING". The caution shall be legible from the distance shown on the sign. Hearing protectors shall not be supplied with the pump units. Plates shall be in accordance with MIL-P-514, type III, composition C of type I, grade A, class 1 material.

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3.7 Reliability. The specified mean-time-between-failure (MTBF) shall be 200 hours when the pump unit is operating at rated capacity and tested as specified in 4.5.2.6.

3.8 Maintainability.

3.8.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, or lubrication charts, as specified (see 6.5 and 6.6), shall be adequate to perform the intended function of assisting or conducting maintenance operations when tested as specified in 4.5.2.7.1.

3.8.2 Maintenance ratio. The pump unit shall have a maintenance ratio of not more than 0.07. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing.

3.9 Pump.

3.9.1 Performance. The pump shall operate as specified herein at any continuous governed engine speed from 1200 revolutions per minute (rpm) to 2100 rpm. When operated with the two stages in parallel, the pump shall deliver 900 gallons per minute (gpm) of liquid petroleum fuel against a total head of 275 feet, and 2000 gpm against a total head of 225 feet. When operated with the two stages in series, the pump shall deliver 500 gpm of liquid petroleum fuel against a total head of 555 feet, and 1400 gpm against a total head of 275 feet. Pump performance shall be based upon pumping liquid petroleum fuel having a specific gravity of 0.80, operation at 2100 rpm, and a static suction lift equivalent to zero feet at sea level atmospheric conditions (29.92 inches of mercury barometric pressure). The pump shall deliver its rated capacity for a period of 30 minutes when positioned at an angle of 15 degrees from the horizontal in any plane of inclination without malfunctioning or misalinement of any component. The pump shall develop a peak efficiency of not less than 78 percent in both series and parallel conditions.

3.9.2 Pump shaft. The pump shaft shall be corrosion-resisting steel conforming to ASTM A276, type UNS S 41000, condition H, or ASTM A582, type S 41600, condition H, or steel conforming to AISI 4140, cold finished, with an ultimate tensile strength of not less than 130,000 pounds per square inch (psi) and a Brinell hardness range of not less than 285 nor more than 310. When the pump shaft is not made of corrosion-resisting steel, a pump shaft sleeve for mounting the shaft seal shall be provided. The sleeve shall be corrosion-resisting steel conforming to ASTM A276, type 304. All corrosion resisting steel shall be passivated in-accordance with QQ-P-35.

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The check valve shall be removable from the pump by removing

3.9.3 Hydrostatic pressure. The pump case shall withstand a 1000 psi internal hydrostatic pressure without leakage through the pump case or joints. The pump shaft seals shall not leak in excess of one drop per minute when subjected to a 1000 psi internal hydrostatic pressure.

3.9.4 Fuel tank. A fuel tank shall be furnished. The capacity of the tank shall be sufficient for not less than 4 hours of continuous operation at the rated load. The tank shall be equipped with a cap and strainer. The tank shall be provided with a hand-operated fuel shutoff valve attached directly to the tank and a mechanically actuated fuel gage mounted on the tank. The tank outlet or the fuel line shall be provided with a 3-way selector valve and a 1/2-inch x 20 SAE flared fitting for connection with an auxiliary fuel supply. A corrosion-resisting circular metal face plate shall be mounted on the fuel selector valve and shall be marked "OFF", "UNIT TANK" and "AUXILIARY".

3.9.4.1 Fuel tank auxiliary control. The fuel tank shall be furnished with an automatic float-type fuel tank auxiliary control which shall control the flow of fuel into the tank. The filler device shall operate on supply line pressures of 1 to 25 psi.

3.10 Line pressure safety switch system. The pump unit shall be equipped with a safety switch system for immediate and positive throttling of the engine to idling speed. The system shall consist of the following components: a hermetically sealed pressure-actuated electric high-pressure switch, a hermetically sealed pressure-actuated electric low-pressure switch, a solenoid, a toggle switch with indicating light, and an independent 12-volt electric circuit. The safety switches shall be integrated with the engine governing system and shall control the action of the engine fuel injection pump. Any one or combination of the following high- and low-pressure conditions shall cause the safety-switch system to function:

a. A drop in pressure below the present limits when the flow of liquid in the upstream (suction) line is cut off, leaving the pump to operate dry.

b. A drop in pressure below the preset limits when the liquid is being wasted because of a break in the downstream (discharge) line.

c. A rise in pressure beyond the preset limits due to a block in the downstream (discharge) line.

3.10.1 Pressure-actuated switches. The high- and low-pressure safety switches shall be exterior surface-mounted bellows-type, pressure-actuated switches with positive quick-make-and-break action. The switches shall be in the open position when the pump is operating. The operating range of the switches shall be 90 to 900 psi, plus or minus 5 percent. The switches shall be adjustable for any set point within their operating range without breaking the hermetic seal. The switches shall be connected to the pump discharge nozzle with copper tubing and fittings.

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3.10.2 Solenoid. A weatherproof pull-type manual-resetting solenoid shall be securely mounted on the engine with the plunger arm connected to the diesel fuel injection pump. The solenoid shall be connected by electric cables to the rest of the control system. When energized, the plunger arm shall act to have the engine operate at idling speed. Provisions shall be provided for adjusting the engine speed to the engine manufacturer's recommended idling speed when the solenoid is energized.

3.10.3 Electric circuit. The wiring shall be arranged so that operation of the toggle switch will open or close the circuit to the solenoid. With the toggle switch in the "on" position, the circuit shall be closed and the indicating green light shall be energized. With the toggle switch in the "off" position, the circuit shall be open and the indicating light shall be off. The circuit shall be such that the toggle switch must be in the "on" position for the pressure-actuated safety switch system to function. When the pump is put into operation with the line pressure below the preset limits, the opening of the toggle switch shall open the circuit to the pressure switches to allow engine starting.

3.10.4 Electronic pressure control. When specified (see 6.2) an electronic pressure control system employing solid state components shall be substituted for the "line pressure safety switch system" and all functions described in 3.10 will be performed.

3.11 Pressure gages. Two pressure gages of the Bourdon-tube type shall be provided. Each gage shall have a range from 0 to 1000 psi and shall be accurate to within 3 percent of the total gage reading. The dials shall be not less than 5 inches in diameter and shall be marked in red from 600 to 1000 psi.

3.11.1 Gage savers. Gage savers shall be provided on the lines connecting the pressure gages and safety switches to the pump to protect them from damage caused by freezing or pressure surging. The device or devices used for pressure-dampening liquid barriers and the isolated liquid between the barriers and the gages shall assure required operation of the instruments under all operating conditions of the pump.

3.12 Performance data chart. A performance data chart approximately 8-1/2 x 11 inches shall be furnished. The chart shall show the complete performance characteristics of the pump for both parallel and series operation when operating at 1850 rpm and 2100 rpm and pumping liquid petroleum fuel having a specific gravity of 0.80. All data shall be derived from the actual performance of the first article pump unit. The chart shall be a photosensitive anodized aluminum plate conforming to MIL-P-514. The chart shall be located on the inside of the instrument panel cover and shall be riveted to the cover.

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3.13 Engine. The diesel engine shall be a heavy-duty commercial industrial type, with replaceable cylinder liners, capable of operating on diesel fuel conforming to VV-F-800 and lubricating oil conforming to MIL-L-2104 and MIL-L-46167. The average oil consumption during first article testing shall not exceed .0035 pounds per brake horsepower hour. Full-flow-type oil filters shall be utilized in the engine lubrication system. The temperature of the oil in the engine oil gallery shall not exceed 230° F when tested as specified herein. When an oil cooler is provided, the temperature of the oil entering the cooler or in the engine sump shall not exceed 250° F when tested as specified herein. The temperature of the coolant in the top tank of the radiator shall not exceed 210° F when tested as specified herein. Primary and secondary fuel filters shall be furnished and may be of the spin-on type. The engine's electrical system shall be 12 volts and shall be equipped with an alternator of not less than 35 ampere capacity. The engine shall be equipped with the following:

a. Engine lubricating oil temperature gage.
b. Engine coolant temperature gage.
c. Tachometer in accordance with SAE J678d.
d. Battery generator ammeter.
e. Hourmeter, with time-totalizing mechanism of not less than 9999 hours, in accordance with SAE J678d.
f. Fuel level indicator.
g. Safety control to shut down engine when coolant temperature reaches 220 ° F.
h. Safety control to shut down engine when engine oil pressure drops below 15 psi.
i. Safety control to shut down engine when the engine speed exceeds engine manufacturer's recommended maximum speed.

Note: All instruments and controls shall be mounted on the pump control panel.

j. A weatherproof battery box painted with acid-resistant paint.
k. Manual engine control of the push-pull locking micrometer-type throttle that shall provide for operating the engine at any speed from idle to maximum governed speed.
l. A muffler and exhaust pipe located to discharge the exhaust gases away from the operator. The exhaust pipe shall be fitted with a raincap in accordance with MS51389 or shall be designed to preclude the entry of blowing rain and snow.
m. A steel housing complete with side panels.
n. A lifting eye attached to the engine block or head at the balance point of the unit. The lifting eye shall extend through the engine housing hood for easy access for hookup.
o. Spark arrester in accordance with Forest Service Standard 5100-1A. The spark arrester and muffler (item l) may be a combined unit.

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3.13.1 Engine housing. The engine shall be provided with a housing that shall enclose the engine sides, top, and bottom. The housing shall incorporate the following: instrument panel with cover, removable side panels, winterfront, removable bottom panel, and weather stripping. The housing shall be constructed of steel with a minimum nominal thickness of 0.047 inch. Drip channels or other arrangements shall be provided to prevent the entry of rain or melted snow.

3.13.1.1 Instrument panel. The instrument panel shall be located on the pump end of the housing in a separate compartment flush with the outside housing. The instrument panel shall be provided with an interior switch-actuated panel light and vertically hinged cover incorporating continuous (piano) type hinges, a latch requiring manual action for closing, and provisions for holding the cover in the open position.

3.13.1.2 Removable side panels. The removable side panels shall be provided with hinged ventilation doors to enable the engine to operate as specified herein with the side panels in place. When open, the ventilation doors shall prevent the entry of rain or snow into the housing. The side panels shall also be provided with access doors large enough to permit an operator wearing arctic mittens to check the crankcase oil level and to reset the safety switch solenoid. Both the ventilation and access doors shall be provided with latches and provisions for holding them in the open position. The side panels shall be provided with manual action flush-mounted latches to hold them in place and with hasps to permit locking the panels.

3.13.1.3 Winterfront. A metal winterfront shall be furnished for the radiator. The front shall be either a shutter, louver, or sliding-plate type and shall be thermostatically controlled with provisions for auxiliary manual control. The manual control shall bypass the thermostat control by means of a detent, a spring, or a slotted cam arrangement. The winterfront shall be protected by an expanded metal guard grille conforming to MIL-M-17194, type I or type II, class 2, grade A.

3.13.1.4 Bottom panel. A removable botton panel shall be provided between the front and rear engine supports. The panel shall have a drainhole with a removable cover and shall be located approximately at the top of the skid base. The panel shall be removable through the opening provided when the side panels are removed.

3.13.1.5 Heat shield. Heat shields shall be provided wherever necessary to prevent damage to or warpage of the engine housing and instrument panel.

3.13.1.6 Housing weather stripping. Flexible double-lip, extruded-sectional silicon elastomer material shall be used for weather stripping between the housing and the doors and covers. The weather stripping shall seal uniformly and continuously around the entire opening of each door and cover.

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3.13.2 Power and speed rating. The power and speed rating of the engine shall be such that operation of the pump, as specified in 3.9.1 and any other operating condition specified herein, at an engine speed of 2100 rpm while pumping liquid petroleum fuel having a specific gravity of 0.80 shall not require horsepower in excess of the continuous duty rating of the engine as previously published by the engine manufacturer or 85 percent of the engine manufacturer's previously published maximum horsepower rating, at the applied speed, whichever is less.

3.14 Electromagnetic interference. The electromagnetic interference emission characteristics of the pump unit shall conform to MIL-STD-461, for engine driven equipment.

3.15 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.15.1 Lubricants. The pump unit shall operate as specified herein when lubricated with military lubricants (see 6.7).

3.15.2 Oil lubrication. Unless otherwise specified herein, all components shall be filled to the operating level with military oils designated for use in the temperature range specified herein. A tag shall be attached in a conspicuous place to indicate the temperature range and grade of lubricant used.

3.16 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition I or II, as applicable (see 3.16.1), with 1 percent copper 8-quniolinolate or 6 percent salicylanilide (by weight) based on the nonvolatile content of the varnish:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.

b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

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3.16.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of Composition I. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 102, South Coast Air Quality Management District, County of Los Angeles, CA.

3.17 Toolbox. The pump unit shall be provided with a weatherproof toolbox made of 14 gauge sheet steel. The toolbox shall hold the publications and operating and maintenance tools (see 6.5 and 6.6). The toolbox shall have an overlapping hinged lid to prevent the entry of rain or dirt. The lid shall be mounted on hinges conforming to MS35829 with a 0.125-inch-diameter brass pin. When the toolbox is mounted on the pump, the toolbox lid shall open to a minimum of 90 degrees from the horizontal and remain in the open position. The toolbox shall be provided with a latch that will secure the cover in the closed position. In addition, the toolbox shall be provided with a hasp for locking the box. The latch and hasp may be either one combined unit or two separate units. Two drainholes in opposite corners of the bottom of the box shall be provided. The toolbox shall be fastened to the pump skid base in a protected location.

3.18 Identification marking. The pump units shall be identified in accordance with MIL-STD-130. The marking shall be applied to the pump unit on plates conforming to MIL-P-514, type I, style I, composition C, of type I, grade A, class 1 material. Each plate shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.19 Instruction plates. Each pump unit shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the pump unit. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. Each plate shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.20 Treatment and painting. The portions of the pumping assembly normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A.

3.21 Workmanship.

3.21.1 Castings and forgings. All parts, components, and assemblies of the pump unit which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

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[illegible] operating the pump unit at rated speed and at 900 gpm at a suction head

3.21.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free from burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not deterimental to safety.

3.21.3 Welders and welding.

3.21.3.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/AWS D1.1, Structural Welding Code, Section 5, Qualification.
ANSI/ASME, Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with the ANSI/AWS or ANSI/ASME Code.

3.21.3.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the pump unit to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the pump unit shall be at the option of the contractor.

3.21.4 Bolted connections. Bolt holes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than two full threads.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insurng that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.1.1 Engine test. Prior to installation of the engine in the pump unit and examination and test of the first article, the engine shall be tested in accordance with MIL-STD-1400, test method series 2000, except test method 2400 shall not apply. Upon completion of this test and prior to installation of the engine in the first article, the fuel system and governor shall be sealed by a Government representative at the settings established by this test. Seals shall also be installed to prevent separation of the oil pan, the cylinder head, and the valve covers from the cylinder block. Any changes made to these settings or removal or alteration of the seals or adjustments by other than a Government representative or without his expressed consent shall be cause for rejection.

4.1.1.2 Production engine tests. Each engine shall be tested in accordance with MIL-STD-1400, test method series 4000, except that the engine manufacturer's standard production test may be substituted provided it equals or exceeds test method series 4000.

4.1.1.3 Production control test. One engine in each lot (lot size not to exceed 50 engines) shall be examined and tested in accordance with MIL-STD-1400, test method series 3000.

4.1.2 Disassembly inspection. Failure of any examination or test by the first article shall be cause for disassembly, in the presence of a Government representative, of the first article to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

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4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.8 and 6.9.1).

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 Examination. The first article pump unit shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests. The first article pump unit shall be subjected to the tests marked "X" in column 1 of table I. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests.

4.4.3.1 Individual. Each pump unit shall be subjected to the tests marked "X" in column 2 of table I. Failure of any test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be subjected to the tests marked "X" in column 3 of table I. AQL shall be 4.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The pumping assembly shall be examined as specified herein for the following defects:

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Major

101. Any part or component not in accordance with the QAP requirements as shown on the drawings.
102. Material not as specified.
103. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment.
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
105. Contractor does not have documentation available for identification of material, material finishes or treatments.
106. Components missing or not as specified.
107. Parts or components do not function as specified.
108. Electromagnetic interference suppression components and network not identical to the approved model.
109. Workmanship not as specified.

Minor

201. Any dimensions, other than those identified on the QAP requirements shown on the drawings, not as specified.
202. Data charts not as specified or missing.
203. Treatment and painting not as specified.
204. Identification marking missing or not legible.
205. Instruction plates not legible.

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed in accordance with the applicable provisions of the test code for centrifugal pumps of the standards of the Hydraulic Institute. All performance data shall be corrected to standard sea level conditions of barometric pressure of 29.92 inches of mercury and liquid fuel at 60° F having a specific gravity of 0.75 and vapor pressure of 4.0 psi. Water at a temperature of between 60° F and 80° F may be used as the test fluid.

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4.5.2.1.1 Test schedule.

Table I. Test schedule

Quality conformance			Tests	Test paragraph	Requirement paragraph
First article	Individual	Sample			
1	2	3	4	5	6
X	-	-	Engine	4.1.1.1	3.13
-	X	-	Production	4.1.1.2	3.13
-	-	X	Production control	4.1.1.3	3.13
X	-	-	High temperature	4.5.2.2	3.4
X	-	-	Low temperature	4.5.2.3	3.4
X	-	-	Safety	4.5.2.4	3.5
X	-	-	Human factors	4.5.2.5	3.6
X	-	-	Reliability	4.5.2.6	3.7
X	-	-	Maintenance support evaluation	4.5.2.7.1	3.8.1
X	-	-	Maintenance ratio	4.5.2.7.2	3.8.2
X	-	-	Pumping	4.5.2.8.1	3.9.1
X	-	-	Operational (24-hour)	4.5.2.8.2	3.9.1
X	-	X	Production	4.5.2.8.3	3.9.1
X	-	X	Shaft material	4.5.2.9	3.9.2
X	X	-	Pump case	4.5.2.10.1	3.9.3
X	X	-	Fuel tank	4.5.2.10.2	3.9.4
X	-	X	Line pressure safety-switch system	4.5.2.11	3.10
X	-	-	Tilted position	4.5.2.12	3.9.1
X	-	-	Lifting bail	4.5.2.13	3.9.1
X	-	-	Electromagnetic interterference	4.5.2.14	3.14

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4.5.2.2 High temperature. The pump unit shall be tested as specified in MIL-STD-810, procedure I, method 501. The maximum ambient storage temperature shall be 145° F, and the maximum ambient operating temperature shall be 110° F. The operating period of procedure I shall be 2 hours. The pump unit shall be operated at rated capacity. Nonconformance to 3.4 shall constitute failure of this test.

4.5.2.3 Low temperature. The pump unit shall be tested as specified in MIL-STD-810, procedure I, method 501. The minimum ambient storage temperature shall be minus 28° F. The minimum ambient operating temperature shall be minus 25° F. The operating period of procedure I shall be 2 hours. The pump unit shall be operated at rated capacity. Nonconformance to 3.4 shall constitute failure of this test.

4.5.2.4 Safety. The pump unit shall be evaluated throughout testing as specified in 4.5.2. Nonconformance to 3.5 shall constitute failure of this test.

4.5.2.5 Human factors. The human factors shall be evaluated during testing. Nonconformance to 3.6 shall constitute failure of this test.

4.5.2.5.1 Noise level measurement.

4.5.2.5.1.1 Representative duty position. Measure the noise level at every 45 degrees ± 5 degrees of azimuth, 65 inches ± 1 inch above ground level, at a 25-foot radius from the geometric center of the equipment in accordance with MIL-STD-1474 for steady-state noise. The pump unit shall be operated at rated load. Nonconformance to 3.6.1 shall constitute failure of this test.

4.5.2.5.1.2 Exterior. Measure the noise level in accordance with MIL-STD-1474 for exterior noise. Nonconformance to 3.6.1 shall constitute failure of this test.

4.5.2.5.1.3 Hazard precautions. Measure the noise level at all control and maintenance positions while operating at the maximum noise condition for those locations. Should the noise level exceed 85 dB(A) at any position, failure to provide the precautions specified in 3.6.1 shall constitute failure of this test.

4.5.2.6 Reliability. Using the MTBF specified in 3.7 the pump unit shall be tested as specified in 4.5.2.8.2 with "accept" and "reject" criteria in accordance with figure 1. Continue testing until either an "accept" or "reject" decision is reached. A failure is defined as any malfunction which cannot be corrected within 30 minutes (all waiting periods are excluded) by adjustment, repair, or replacement action using controls and on-equipment tools or parts available to the using element, and is a malfunction which causes or may cause one or more of the following:

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a. Failure to commence operation, cessation of operation, or degradation of performance below specified levels.
b. Damage to the pump unit by continued operation.
c. Safety hazards to personnel.

Nonconformance to 3.7 shall constitute failure of this test.

4.5.2.7 Maintainability.

4.5.2.7.1 Maintenance support evaluation. Maintenance operations required during first article testing shall be accomplished to determine conformance to 3.8.1. Failure of the support elements to maintain the pump unit as specified shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio or the MTBF.

4.5.2.7.2 Maintenance ratio. The maintenance ratio shall be computed during first article testing. Nonconformance to 3.8.2 shall constitute failure of this test.

4.5.2.8 Performance.

4.5.2.8.1 Pumping. Operate the pump as specified herein in both the series and parallel conditions to determine brake horsepower, efficiency, and performance. Operate the pump at zero feet equivalent suction lift for both series and parallel operations. Operate the pump between shutoff conditions and 1500 gpm for series operation and between shutoff conditions and 3000 gpm for parallel operation. Plot performance curves showing gallons per minute versus total dynamic head, brake horsepower required, pump efficiency, and the net positive suction head required from the data obtained. The test shall be conducted in accordance with the Rating Standards and Test Code Standards of the Hydraulic Institute. Performance at rates less than those specified in 3.9.1 shall constitute failure of this test.

4.5.2.8.2 Operational (24 hour). Operate the pump unit in the parallel connection at 2000 gpm against a total head of 225 feet at 2100 rpm for 8 hours continuously. Change the pump unit to the series connection and operate at 1400 gpm against a total head of 275 feet at 2100 rpm for 16 hours continuously. Examine the pump unit at the end of each 8-hour period during the 24 hours. Maintenance and minor adjustments shall be permitted during the examination periods. Examine the pump unit during operation for leakage through the pump casing or the shaft seals and for malfunction of any component. Evidence of leakage, malfunction of any component, inability of the pump to deliver the minimum capacities specified herein, or any damage to the engine housing due to heat buildup from either the engine or the exhaust system shall constitute failure of this test.

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4.5.2.8.3 Production. Test the pump unit in accordance with the Rating Standards and Test Code Standards of the Hydraulic Institute for capacity at 2100 rpm and a total head of 555 feet. At the conclusion of this test, operate the unit for not less than 30 minutes at the above condition. Inability of the pump to deliver a minimum of 500 gpm, leakage through the pump case, joints, or shaft seal, or malfunction of any component shall constitute failure of this test.

4.5.2.9 Shaft material. One sample from each batch of steel used in the manufacture of pump shafts shall be tested for tensile strength and Brinell hardness. An ultimate tensile strength of less than 130,000 psi or a Brinell hardness of less than 285 or more than 310 shall constitute failure of this test.

4.5.2.10 Pressure.

4.5.2.10.1 Pump case. Apply an internal hydrostatic pressure of 1000 psi for a period of not less than 10 minutes to the completely assembled pump with seals in place. A leakage in excess of one drop per minute at each shaft seal or any leakage through the pump case or joints shall constitute failure of this test.

4.5.2.10.2 Fuel tank. Apply an internal air pressure of 5 psig to the fuel tank with all attachments and immerse in water so that the tank top is not less than 12 inches under the water level. Any evidence of leakage from the tank shall constitute failure of this test.

4.5.2.11 Line pressure safety switch system. During the tests specified in 4.5.2.8.2, simulate high- and low-pressure conditions by restricting the pump suction and discharge not less than two times during each 8-hour test period. The low-pressure switch shall be set at 90 psi and the high-pressure switch at 110 psi for parallel operation; the settings shall be 110 psi and 130 psi, respectively, for series operation. During the test specified in 4.5.2.8.3, the settings shall be 230 psi and 250 psi, respectively. The test shall be performed one time. Inability of the safety switch system to reduce the engine speed to idling when the simulated high- and low-pressure conditions exceed the safety switch settings shall constitute failure of this test.

4.5.2.12 Tilted position. Operate the pump unit at its rated capacity for not less than 30 minutes when positioned 15 degrees from level along the longitudinal centerline of the skid base. Evidence of malfunction or misalinement of components shall constitute failure of this test.

4.5.2.13 Lifting bail. Place the pump unit on a flat level surface and lift it with the lifting bail. Raise the pump unit until the skid has completely cleared the ground. With the pump unit in this position, measure the difference in elevation between the high and low ends of the skid. A difference in elevation greater than 12 inches shall constitute failure of this test.

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4.5.2.14 Electromagnetic interference. The first article unit shall be tested to determine conformance with 3.14. The contractor shall furnish the contracting officer a report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.6 Inspection comparison. The Government may select pump units at any time during the contract production period and subject these pump units to the examination and the tests specified in 4.3 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on pump units selected at random from those which have been accepted by the Government and will not include the previously inspected first article pump units. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison pump unit to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison pump unit and shall be cause for refusal by the Government to continue acceptance of production pump units until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on pump units previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all pump units accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging.

4.7.1 Preproduction pack inspection.

4.7.1.1 Examination. The preproduction pack shall be examined for defects specified in 4.7.2.3. Presence of one or more defects shall be cause for rejection.

4.7.1.2 Test. The preproduction pack of the crated pump unit for level A or B shall be subjected to the rail-impact test as specified in MIL-STD-1186, appendix A. At the conclusion of the preproduction pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of hold-downs, ties, stays, blocking or bracing, or any visual damage to the components shall constitute failure of the preproduction pack and shall be cause for rejection.

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4.7.2 Quality conformance inspection of pack.

4.7.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.2.3 Examination. Samples selected in accordance with 4.7.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

110.	Materials, methods, or containers not as specified for level A or B. Each incorrect material, method, or container shall be considered one defect.
111.	Disassembly not as specified for level A.
112.	Matchmarking not as specified for level A.
113.	Matchmarking information not applied with waterproof ink as specified for level A.
114.	Unprotected surfaces not coated with preservative as specified for level A.
115.	Engine not preserved in accordance with the referenced document and exception as specified for level A.
116.	Pump not preserved as specified for level A.
117.	Pump openings not sealed as specified for level A.
118.	Information tag indicating pump is filled with preservation oil not attached as specified for level A.
119.	Crossover fittings and Y-type fittings not removed or the surfaces not preserved as specified for level A.
120.	Flexible couplings not preserved as specified for level A.
121.	Repair parts not preserved in accordance with the referenced document for level A.
122.	Tools not preserved in accordance with the referenced document for level A.
123.	Consolidation not as specified for level A.
124.	Preservation and packing not in accordance with the referenced document as specified for commercial.
125.	Engine and engine accessories not preserved as specified for commercial.
126.	Packing not as specified for level A or B.
127.	Marking missing, illegible, incorrect, or incomplete for level A, B, or commercial.

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5. PACKAGING

5.1 Preproduction pack. The contractor shall furnish a preproduction pack for examination and testing of the crated pump unit for level A or B within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination and test shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The preproduction pack may be accomplished utilizing either the first article pump unit or the production pump unit. If the first article pump unit is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government, to facilitate comparison between the first article pump unit and the production pump unit.

5.2 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Disassembly and matchmarking.

5.2.1.1.1 Disassembly. Disassembly of the pump unit shall be limited to the removal of protruding parts that tend to increase cubage. Bolts, nuts, screws, pins, and washers removed shall be reinstalled and secured in mating parts to prevent their loss.

5.2.1.1.2 Matchmarking. Removed parts and mating parts shall be matchmarked to facilitate reassembly. Matchmarking identification shall be on cloth shipping tags conforming to UU-T-81, type A, and the tags shall be attached to mating parts. The information on the tags shall be applied with waterproof ink.

5.2.1.2 Preservatives. Preservatives specified herein by "P" numbers shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.2.1.3 Unprotected surfaces. Unpainted exterior metal surfaces of components of the pump unit requirimg the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative.

5.2.1.4 Engine. The engine and engine accessories shall be preserved in accordance with the level A requirements of MIL-E-10062 type II, method I, except the engine cooling system shall be preserved by the water and antifreeze procedure using antifreeze conforming to MIL-A-46153.

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5.2.1.5 Pump. The interior surfaces of the pump shall be coated with type P-10, grade 30 lubricating oil. Preservation of the pump after assembly and for standby storage shall be accomplished as follows. The pump shall be drained, and the drain plugs shall be coated with type P-10, grade 30 lubricating oil before the plugs are reinstalled. The suction and discharge ports shall be closed in a manner to insure an oiltight seal. The air-vent valves shall be removed from the top of the pump, and not less than 1 gallon of type P-10, grade 30 lubricating oil shall be poured through each opening. The valves shall be reinstalled. The oil plugs over the shaft seals shall be removed and type P-10, grade 30 lubricating oil shall be poured through each opening to apply lubricating oil to the carbon inserts. The oil plugs shall be coated with type P-10, grade 30 lubricating oil and reinstalled. The pump shall be rotated with the engine starter for approximately 10 seconds and, after 1-minute standby, again rotated to insure complete coverage of all interior surfaces of the pump. The oil shall be left in the pump for shipment and storage. A tag conforming to UU-T-81, type A shall be attached indicating: "The pump is filled with preservative lubricating oil. Drain and refill with the proper lubricant before placing the pump in operation". The information on the tag shall be applied with waterproof ink.

5.2.1.5.1 Pump openings. The suction and discharge openings shall be covered with barrier material conforming to MIL-B-121, type II, grade A, class 1, secured in place with tape conforming to MIL-T-22085, type II.

5.2.1.5.2 Crossover and Y-type fittings. The crossover and Y-type fittings shall be removed from the pump. Unpainted exterior surfaces shall be coated with type P-1 preservative. Interior surfaces shall be coated with type P-10, grade 30 lubricating oil. Excess preservative shall be drained, and the openings shall be covered with barrier material conforming to MIL-B-121, type II, grade A, class 1, secured in place with tape conforming to MIL-T-22085, type II.

5.2.1.6 Flexible coupling. The coupling flexible elements shall be removed from the coupling, coated with type P-10, grade 30 lubricating oil and preserved together in accordance with MIL-P-116, method IC-1. The preserved flexible elements shall be consolidated in accordance with 5.2.1.10.

5.2.1.7 Repair parts. Repair parts for each pump unit shall be preserved in accordance with the level A requirements of MIL-R-196. Parts not specifically covered in MIL-R-196 shall be preserved in accordance with the applicability, selection and application provisions of MIL-P-116.

5.2.1.8 Maintenance tools. Maintenance tools shall be preserved in accordance with the level A preservation and packaging requirements of PPP-P-40.

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5.2.1.9 Technical publications. Technical publications for each pump unit shall be preserved together in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.1.10 Consolidation. The repair parts, tools, technical publications, and components removed by disassembly shall be consolidated in the toolbox as space permits. Items not fitting in the toolbox shall be consolidated in boxes conforming to PPP-B-601, domestic type, style optional or PPP-B-621, class 1, style optional. Cushioning or blocking as applicable shall be provided to prevent movement or damage. The toolbox lid shall be closed and secured. Strapping of the boxes shall not be required.

5.2.2 Commercial. Each complete pump unit with repair parts, tools, and publications, except for the engine and engine accessories (see 5.2.2.1) shall be preserved in accordance with MIL-STD-1188.

5.2.2.1 Engine. The liquid cooling system shall be filled with a fresh clean solution of 50 percent water and 50 percent antifreeze conforming to MIL-A-46153. A tag shall be attached indicating the temperature to which the cooling system can be subjected before it is damaged. All lubrication reservoirs shall be filled with the lubricant or fluid designated for use in the temperature range at the shipping destination. A tag or tags, as applicable, shall be attached in a conspicuous location indicating which lubricant or fluid has been used.

5.3 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.3.1 Level A. Each complete pump unit, preserved as specified in 5.2, shall be packed in a crate conforming to MIL-C-104, type I, class 1 or 2, style A. Blocking, bracing, anchoring, and strapping shall be in accordance with the crate specification and in accordance with MIL-STD-1186.

5.3.2 Level B. Each complete pump unit preserved as specified in 5.2, shall be packed in a crate conforming to MIL-C-3774, type I, style A. Blocking, bracing, anchoring, and strapping shall be in accordance with the appendix to the crate specification and in accordance with MIL-STD-1186.

5.3.3 Commercial. Each complete pump unit, preserved as specified in 5.2, shall be packed in accordance with MIL-STD-1188.

5.4 Marking.

5.4.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.4.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

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6. NOTES

6.1 Intended use. The pump unit is intended for use as a trunkline pump when operated with the stages in series and as a booster pump when operated with the stages in parallel.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required for inspection and approval, and number of pump units required (see 3.2).

c. When the Government will conduct any or all of the first article pump examinations and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).

d. When electronic pressure control system shall be used in lieu of line pressure safety switch system (see 3.10.4).

e. When the welding process will not be at the option of the contractor (see 3.21.3.2).

f. Time frame required for submission of preproduction pack (see 5.1).

g. Degree of preservation and degree of packing required (see 5.2 and 5.3).

6.3 First article. When a first article inspection is required, the item will be tested and should be a preproduction model pump unit. The preproduction model should consist of one or more units. The contracting officer should include specific instruction in acquisition documents regarding arrangements for examinations, test, and approval of the first article.

6.4 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the pump units in accordance with this specification.

6.5 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each pump unit.

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6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools, as necessary (including any special tools), and instructions regarding shipment of pump units.

6.7 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the pump unit covered by this specification as contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.8 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

> "The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.9 Definitions.

6.9.1 Quality assurance provisions(QAP). A QAP is a contractual requirement that supplements section 4 of the specification. A QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.10 Noise limits. When the noise level of the pump unit exceeds 85 dB(A), the appropriate discussion of the noise hazard in accordance with MIL-STD-1474 (MANUALS Paragraph) shall include the requirement for hearing protection, the noise level of the pump unit, the distance from the pump unit at which 85 dB(A) will always be met, and what operational situation or system configuration will reduce the level to 85 dB(A).

6.11 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Custodians:
Army - ME
Air Force - 99

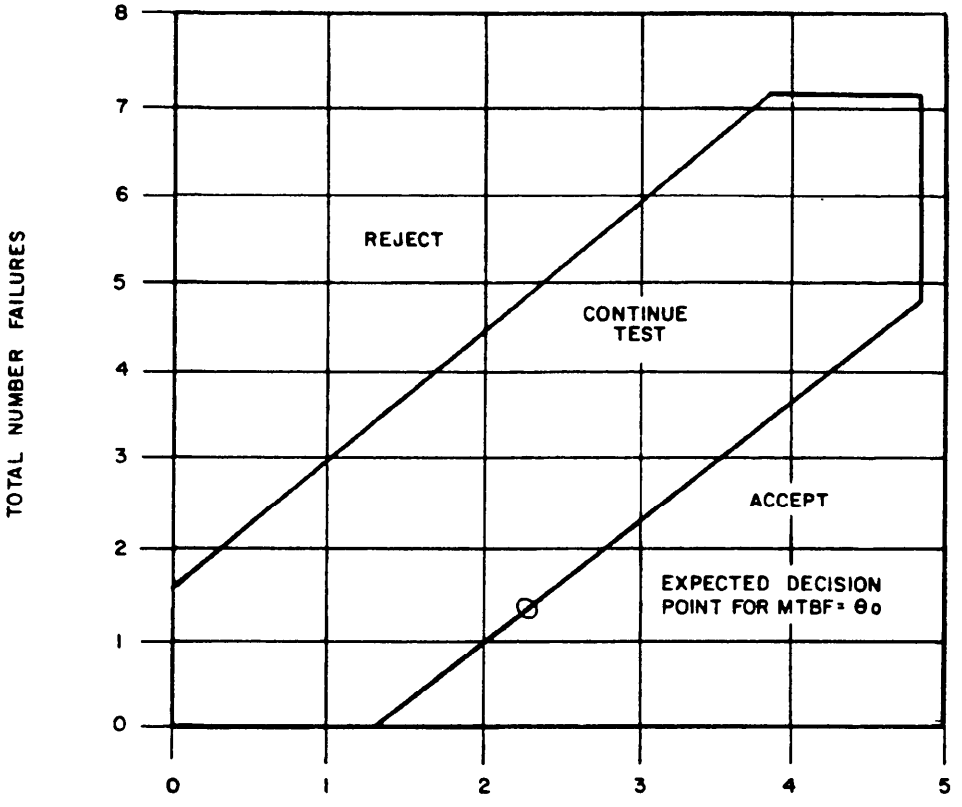
Review activity:
Air Force - 84

Preparing activity:
Army - ME

Project 4320-0289

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DECISION RISKS 20%
DISCRIMINATION RATIO 2.0:1



NO. OF FAILURES	TOTAL TEST TIME * REJECT (EQUAL OR LESS)	ACCEPT EQUAL OR MORE
0	N/A	1.40
1	N/A	2.09
2	.35	2.79
3	1.04	3.48
4	1.73	4.17
5	2.43	4.87
6	3.12	4.87
7	3.81	4.87
8	4.87	N/A

*TOTAL TEST TIME IS TOTAL UNIT HOURS OF "EQUIPMENT ON" TIME AND IS EXPRESSED IN MULTIPLES OF THE SPECIFIED MTBF.

FIGURE 1. Accept-reject criteria.

X-3661

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-12620E	Pump, Centrifugal: Petroleum-Pipeline, DED, Skid-Mtd,; 2-Stage, 6-In., 500-GPM at 555-Ft Head to 1,400-GPM at 275

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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